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RESEARCH ARTICLE

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If global or local investor sentiments are prone to developing an impact on stock returns, is there an industry effect?

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Abstract

This article investigates the heterogeneous impacts of either global or local investor sentiments (IS) on stock returns (SR). We study 10 industry sectors through the lens of six (so-called) emerging countries: China, Brazil, India, Mexico, Indonesia, and Turkey, over the 2000 to 2014 period. Using a panel data framework, our study sheds light on a significant effect of local IS on expected returns for basic materials, consumer goods, industrial, and financial industries. Moreover, our results suggest that from global IS alone, one cannot predict expected SR in these markets.

KEYWORDS

emerging countries, global investor sentiments, industry sectors, local investor sentiments, stock returns

JEL CLASSIFICATION

G10; C23

1 | INTRODUCTION

To consider investor sentiments (IS), as a persistent vehicle to capture market fluctuations, is a modern research field (Neal & Wheatley, 1998), but had been already considered by Keynes (1936). Fisher and Statman (2000, 2003) even distinguished “Wall Street strategists from individual investors or newsletter writers.” That was followed by Brown and Cliff (2005), Chang, Faff, and Hwang (2011), and Baker, Wurgler, and Yuan (2012). Schmeling (2009) studied IS and stock returns (SR) in 18 industrialized countries and Baker and Wurgler (2006) found that IS have “larger effects on securities whose valuations are highly subjective and difficult to arbitrage.” Since equities from different industries have different characteristics, the impacts of IS should be expected to vary among industries, as promoted by Chen, Chen, and Lee (2013). In fact, empirical and theoretical evidences (Bu & Pi, 2014; Dhesi & Ausloos, 2016) recently show that investors' (irrational or not) emotions, tied to economic or other news, seem to discrepantly sway the future returns of stocks.

Yet to our knowledge there is only a small number of studies, which focus on the correlation between specific industries' expected returns and IS. Beside the remarkable paper by Chen et al. (2013), only a few scholars have regarded emerging markets as the investigated objects for an IS-SR nexus research.

Our study focusses on industry type correlations with both global and local IS in a few emerging markets. Both local IS (LIS) and global IS (GIS) are collected within a panel data set which contains the 2000 to 2014 period for six (so called) emerging countries, and 10 industries: basic materials, consumer goods, consumer services, financials, health care, industrials, oil and gas, technology, telecommunications, and utilities.[1] For each industry, the SR are computed from the data of FTSE global price indexes. We use each country's (thus local) consumer confidence index (CCI) to reflect LIS; we measure the GIS using the State Street Investor Confidence Index (http://www.statestreet.com/ideas/investor-confidence-index.html). We include business cycle as a control variable in our regression framework returns, following McLean and Zhao's (2014) recommendation.

We find that the LIS has a heterogeneous effect on expected SR in different industries. In particular, the empirical results suggest that the expected returns of basic materials, consumer goods, industrials and financials industries have significant and positive correlations with LIS. This finding is consistent with most of previous conclusions. However, we find that GIS do not allow to predict expected SR. Moreover, our conclusive results also suggest that it is inadequate to only use LIS and GIS to perform a rational anticipation of future SR.

Consequently, this study contributes to the literature in three folds. First, the study focuses on the role of industry in the correlation between IS and expected SR. One of our main motivations is to find which industry is the most vulnerable to investors' emotion fluctuations and subsequently which industry is most stable during the bullish or bearish sentiment periods. Second, we take both GIS and LIS into consideration. Third, this article provides studies through the lens of emerging countries: China, Brazil, India, Mexico, Indonesia, and Turkey. We use the data between January 2000 and December 2014 to guarantee the volatility and practicability of the research. To the best of our knowledge, this is the first study to investigate the correlation between IS and different industries' expected SR with the panel data of emerging countries, on an equal footing.

The remainder of the article is organized as follows. We propose our hypotheses in Section 2. Section 3 outlines the methodology with our comments on relevant variables. Section 4 presents data adjustments. In Section 5, we test our hypotheses and reports the findings. Section 6 serves as a conclusion. There are several Appendices containing "numerical and technical details."

2 | HYPOTHESES DEVELOPMENT

There are numerous studies which focus on the correlation between SR and IS; details of those previous studies are displayed in Table 1. However, since the existing studies of IS and SR are too numerous to be all mentioned, we only list articles which are representative and frequently cited in the literature. According to Table 1, we can conclude that IS are proved to be influential to expected SR in many countries. We therefore formulate our first hypothesis as follows:

Hypothesis 1 LIS has a positive impact on industry SR in emerging countries.

Next, let it be observed that there is a scarcity of evidence on the impact of global sentiments. As noted earlier, it is important to investigate global sentiments along with local sentiments. Chang et al. (2011), p. 36, confirmed the impact of GIS across international markets, with the claim that the LIS effect is "just an empirical manifestation of the global effect."

Therefore, our second hypothesis reads as follows:

Hypothesis 2 GIS has a positive impact on industry SR in emerging countries, with the effect that more accessible markets will be associated with a stronger GIS.

Notice that Baker et al. (2012) found that both global and local market sentiments have a predictive power on industry returns, global sentiment playing a more significant role. Since only a few researchers perform empirical studies on the relationship(s) between sentiments and SR, distinguishing different industries, the research question (RQ) tied to H1 and H2 immediately follows:

RQ: If GIS or LIS are prone to developing an impact on SR, is there an industry effect? – at least in emerging countries and for a considered time interval.

3 | METHODOLOGY AND VARIABLES

We employ a classical linear panel fixed-effect regression model (Baker & Wurgler, 2006; Brown & Cliff, 2005; Chen et al., 2013; Schmeling, 2009; Simpson & Ramchander, 2002; Tsuji, 2006) which will be separately conducted for the different industries in order to make some comparison:

$$SR_{it+1} = \beta_{0i} + \beta_1 LIS_{it} + \beta_2 GIS_t + \beta_3 BC_{it} + \varepsilon_{it+1}$$

where $i = 1, 2, 3, ..., N$; $t = 1, ..., T$; i and t refer to the country number and time, respectively; N is the maximum number of countries so examined. Notice that the error should be considered as taken at the date of the forecast, $t + 1$, not the day before.

TABLE 1 About previous studies on investor sentiments and stock returns

Publish year	Author(s)	Research object(s)	Method	Data	Research objective	Research results
2002	Simpson and Ramchander	Australia	Regression	Monthly data 1985–1999	Investigating how investor sentiments influences the time varying part of discounts and premia of the first Australian fund.	Investor sentiment has huge influence on the change of premia on the first Australia fund.
2004	Chelley-Steeley and Siganos	UK	OLS	Monthly data 1975–2001	Whether macroeconomic variables or market sentiment is more influential to momentum profits.	Macroeconomic variables could affect momentum profits, while market sentiment could also influence the profitability of momentum trading strategy.
2005	Brown and Cliff	U.S.	Regression	Monthly data 1963–2000	Investigating that whether a survey of investor sentiment have the ability to estimate market return over following years.	Future return has a negative relationship with investor sentiment.
2006	Baker and Wurgler	U.S.	Regression	Monthly data 1962–2001	Whether sentiment has the cross-sectional effects on stock returns or not.	Investor sentiment is more likely to influence smaller, young, less stable, less profitable and non-dividend-paying stocks.
2006	Tsuji	Japan	Regression	Monthly data 1994–2001	Investigates investor sentiment's predictability for expected stock return and its properties in Japan.	Investor sentiment has predict power for short-term expected stock return, while there are no “naïve extrapolation” and “salience effect” in Japan.
2009	Canbas and Kandir	Turkey	VAR model	Monthly data 1997–2005	Analyzing the correlation between investor sentiment and stock return.	Investor sentiment might not have the ability to anticipate stock return.
2009	Schmeling	Developed countries.	Regression	Monthly data 1985–2004	Using the international evidence to prove the connection between investor sentiment and stock return.	Investor sentiment could predict the stock return and those countries whose culture encourages investment behavior are more vulnerable to the influence of sentiment.

(Continues)

TABLE 1 (Continued)

Publish year	Author(s)	Research object(s)	Method	Data	Research objective	Research results
2010	Kaplanski and Levy	Aviation disasters	Regression	Daily data 1950–2007	Aviation's influence on stock price.	Less stable firms are more vulnerable to investor sentiment. Investor sentiment is a significant factor in estimating future returns.
2010	Kurov	U.S.	OLS	Monthly data 1990–2004	Investigating monetary policy's impacts on investor sentiment.	In bear markets, investor sentiment and monetary policy decision have a strong correlation.
2011	Yu and Yuan	U.S.	GARCH (1,1)	Monthly data 1963–2004	The correlation between investor sentiment and the mean–variance tradeoff.	In the low-sentiment periods, market's expected extra return positively link with its conditional variance, while in the high-sentiment periods, there is no connection between them.
2012	Stambaugh, Yu, and Yuan	Portfolios	Regression	11 anomalies over Aug. 1965 to Jan. 2008	Effect of short-sale impediments.	Sentiment has no relation to returns on the long legs strategies, but short leg strategies are profitable following high sentiment
2013	Chen, Chen, and Lee	11 selected Asian countries	Regression and panel threshold model	Monthly data 1996–2010	Analyzing the global and local investor sentiment's correlation with industry returns with both linear and non-linear methods.	Both global and local sentiment have asymmetric influences on industry returns and relationship between expected industry return and investor sentiment varies with different sentimental intervals.
2014	Huang, Yang, Yang and Sheng	China	Regression	Monthly data 2005–2013	Investigating the influence of investor sentiment on a certain industry's stock return.	In the current period, investor sentiment has a positive relationship with return of an industry, while they are negatively correlated in one lag period.
2014	Oprea and Brad	Romania	Regression	Monthly data 2002–2011	Investigating the influence of investor sentiment on stock prices.	Investor sentiment has a positive relationship with returns from small stocks but not from large stocks.

Note: This table proposes some information about several previous studies (in chronological order) on the relationship between investor sentiments and stock returns. The selection is made in order to provide perspectives; other papers are discussed in the text.

Let for China, Brazil, India, Mexico, Indonesia, and Turkey, the country number to be 1, 2, 3, 4, 5, and 6, respectively. *SR* represents the expected SR for a specific industry, SR_{it+1} denotes the industry SR for country i at time $t+1$; the explanatory variables include each country's CCI as a proxy for LIS (*LIS*), the state street investor confidence index as a proxy for the *GIS*, and the unemployment rate as a proxy for business cycle (*BC*). Accordingly, LIS_{it} and BC_{it} represent local market sentiments, global sentiments and business cycle factor for country i at time t, GIS_t denotes the GIS at time t, while β_1, β_2, β_3 are the regression coefficients of local sentiments, global sentiments, and business cycle, respectively. They also represent the marginal effects of those explanatory variables on industry SR; ε_{it+1} is the disturbance term for country i at time $t+1$. An independent and identically distributed (I.I.D.) assumption on $\varepsilon_{i\ t+1}$ indicates that the data is expected to be independent, identically distributed. This assumption allows one to model the joint probability of the data as the product of the probability of individual data points. The parameter β_{0i} is a time independent term which allows the possibility of country-specific fixed effects.

3.1 | Investor sentiments

LIS and GIS are the dependent variables throughout the empirical tests. Many scholars, for example, Charoenrook (2003), Lemmon and Portniaguina (2006), and Schmeling (2009), have employed the CCI to perform sentiment studies (Trading economics, 2015). All countries considered in our research have their own official consumer confidence indicator and we have employed each country's official CCI as a sentiment proxy for local sentiments throughout the paper.

The State Street Investor Confidence Index quantitatively measures the (global) IS wherever the world country is. This index is built by estimating the changes in institutional investors' position of risky assets and is based on a specialized research model (State Street, 2015). This index relies on the actual financial trading behaviors of institutional investors and demonstrates traders' real feelings about the expected returns.

Figure 1 depicts the GIS and LIS for the time and countries of interest. Notice visually that the GIS index appears to have a weak correlation with the LIS indexes: the LIS and GIS indicators appear to move discrepantly during the investigated time interval. For example, the GIS index trends decreased in the 2000–2004 period, while almost all LIS index trends increased during that time.

3.2 | Industry stock returns

The dependent variable is the SR of different industry sectors. As company's demand and supply within a given industry are affected by environmental forces and economic performances in much the same way (Bain, 1959; Porter, 1980), their SR are in practice expected to be correlated. We follow the industry classification benchmark of the Financial Times Security Exchange (henceforth FTSE) group, which divides all traded equities into 10 different industry categories: basic materials (*MATS*), consumer goods (*GDS*), consumer services (*SVS*), financials

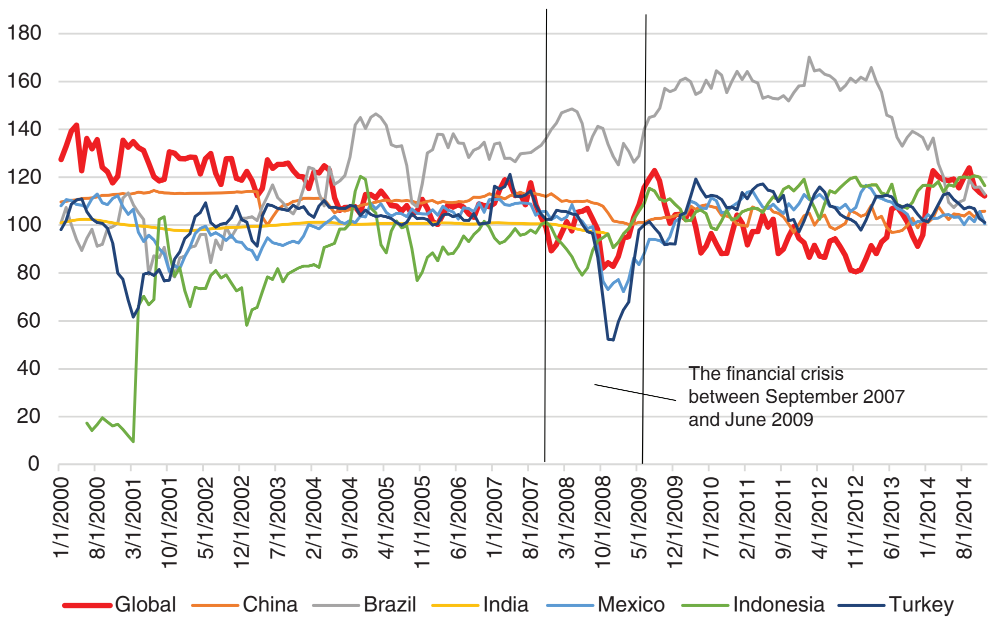

FIGURE 1 Global and local investor sentiments index evolution. The line graph illustrates changes in the global and in six selected countries' investor sentiments indices. The indicators are each country's official consumer confidence index and the State Street Investor Sentiment Index (https://www.investopedia.com/terms/s/state-street-confidence-index.asp). The area between the two black lines represent the financial crisis time between September 2007 and June 2009 [Colour figure can be viewed at wileyonlinelibrary.com]

(*FIN*), health care (*HEA*), industrials (*INDU*), oil and gas (*OIL*), technology (*TECH*), telecommunications (*TELE*), and utilities (*UTIL*). In order to measure each industry's SR, we employ the FTSE "Global Monthly Price Index" (GPI) for each industry sector in the six countries and transform GPIs into log returns. Additionally, the currency unit of the collected price indexes is USD, in order to eliminate the influence of exchange rates. It is fair mentioning that there are some limitations to the FTSE Global Index. First, it only employs some specific portfolios to represent the whole market. Second, the FSTE price index is not available for several countries' specific industry sectors; in our case, for example, there is no FSTE price index for Mexico technology industry.

3.3 | The business cycles

We control the business cycle for macro-economic impacts on SR, using the unemployment rate in each country to capture each country's long-term macro-economic situation. It can be argued that the unemployment rate could only partly reflect a country's long-term macro-economic situation, because it is merely one particular indicator of a country's particular economic condition. Aware of such arguments and limitations, nevertheless, we use the unemployment rate as indicator for measuring the macro-economic status for the practical reason of data availability. One thing needs to be noticed: since the increase of unemployment rate reflects the deterioration of one country's economy, a high unemployment rate therefore denotes negative macro-economic effects. Yet, an increase in unemployment rate might be considered as "good news" by some investors (Boyd, Hu, & Jagannathan, 2005).

4 | DATA AND PRELIMINARY NUMERICAL RESULTS

The data used in this study are obtained from DataStream from January 2001 to December 2014 (DataStream, 2019). According to the Morgan Stanley Capital International Emerging Market Index, 24 developing countries qualify as emerging markets. We first selected the top 7: China, Brazil, India, Russia, Mexico, Indonesia, and Turkey, whose nominal GDP was greater than six hundred million USD in 2014 (International Monetary Fund, 2015). We then had to exclude Russia due to data non-availability, reducing the data comparison to six countries.

It should be emphasized that the raw data, even obtained from validated sources sometimes contain anomalous values. Several industries, such as consumer services (*SVS*) and oil and gas (*OIL*), have maximum returns larger than 100%. In contrast, the minimum return of consumer services (*SVS*) is smaller than −100%. Those phenomena are uncommon for the SR. Of course, the returns of consumer goods (*GDS*), consumer services (*SVS*), health care (*HEA*), oil and gas (*OIL*), and telecommunications (TELE) industries can be significantly negative or positive. Nevertheless, most values do concentrate near zero. Accordingly, unlike the normal distribution, the histograms of those industry returns have relatively high peaks and thin tails. This suggests that data of FTSE global price indexes contains fault values. In order to reduce the influence of anomalous "reported" returns, we first use a widely accepted fraud detection method in order to investigate the problematic aspects of the data set. We then adjusted the data set by deleting all the abnormally repetitious values pointed out when applying Benford's laws (Ausloos, Herteliu, & Ileanu, 2015).

Table 2 reports the adjusted data set's descriptive statistics. Notice (1) almost all of the industries' mean returns are positive except for consumer services industry (*SVS*); this situation partly reflects that industries of the developing world generally enjoy a sustainable growth; (2) the average returns of several industries, for example, the mean return of consumer services (*SVS*), are negative; (3) the average industry returns of basic materials (*MATS*), consumer goods industry (*GDS*), financials (*FIN*), health care (*HEA*), industrials (*INDU*), and industrials (*INDU*) are comparatively higher than the mean industry returns of consumer services (*SVS*), oil and gas (*OIL*), technology (*TECH*), and telecommunications (*TELE*). This is interpreted as follows: the former six industries are the booming industries of the developing world, while the latter four industries are still in an initial development stage; (4) the volatilities of industry returns seem to belong to a normal distribution; this is interpreted as resulting from the eliminations of the extreme values; all the values of standard deviations are close to 5, that is, all the industries' returns fluctuate in "normal ranges"; (5) Table 2 also indicates that the statistical properties of GIS and LIS are distinct from each other. This phenomenon implies our expectations, that is, those two kinds of IS might affect the expected SR in different ways.

Table 3 reports measures of the correlations between the variables. We highlight the following aspects: (a) the correlations between industry returns are all positive but have different extents: some industries' returns (e.g., basic materials (*MATS*)) are closely related to other industries' returns, while several industries' returns (e.g., technology [*TECH*]) have weak correlations with other industry returns; (b) the relationships between industry returns and local sentiments (*GIS*) are all

TABLE 2 Descriptive statistics of the adjusted data set used in this study

Variable	Obs	Mean	Std. Dev.	Min	Max
MATS	1,080	0.2883	5.9418	−35.9707	23.0324
GDS	906	0.4373	4.8700	−21.6857	24.1562
SVS	992	−0.0240	5.5279	−62.0864	19.4596
FIN	1,075	0.3989	5.1834	−26.3445	22.4560
HEA	724	0.2664	4.3248	−17.6841	18.9232
INDUS	1,007	0.2619	5.3034	−34.2964	21.0071
OIL	816	0.0515	5.3762	−28.3126	20.9206
TECH	449	0.0241	5.9233	−25.3684	45.1308
TELE	1,015	0.0987	4.8096	−35.1788	22.9844
UTIL	745	0.3756	4.8605	−19.1293	26.0514
GIS	1,080	109.346	14.120	80.50	141.80
LIS	1,075	106.443	19.028	9.60	170.18
BC	980	6.8956	2.8558	2.15	13.900

Note: This table reports the (rouned) main descriptive statistics of variables after the deletion of abnormal repetitious numbers. The variables are 10 industries' monthly stock return, global investor sentiment, local investor sentiment of six countries, and business cycle factor.

positive but weak; (c) the industry returns of consumer services (*SVS*), financials (*FIN*), health care (*HEA*), industrials (*INDUS*), and oil and gas (*OIL*) are positively linked with the global sentiments (*GIS*), while the other five industries' returns have negative correlations with the global sentiments (*GIS*); (d) both global sentiments (*GIS*) and local sentiments (*LIS*) have a positive but weak correlation; (e) both kinds of IS are positively related to business cycles (*BC*).

5 | EMPIRICAL RESULTS

5.1 | Local IS correlations to the SR

The regression analysis results for the 10 industries are reported in Table 4. The returns of basic materials (*MATS*), consumer goods (*GDS*), financials (*FIN*), and industrials (*INDUS*) have a significant positive correlation with LIS, that is, positive and negative emotions of local traders increase or decrease those industries' expected returns respectively. Furthermore, the coefficients of basic materials (*MATS*) and financials (*FIN*), which are comparative larger than other coefficients, reflect the fact that those two industries have relatively strong reactions to LIS. Besides, the industry returns of consumer services (*SVS*), health care (*HEA*), oil and gas (*OIL*), technology (*TECH*), telecommunications (*TELE*), and utilities (*UTIL*) seem to have no significant relationship with LIS.

Additionally, except for oil and gas (*OIL*), the other five industries' expected returns are insignificantly connected with both GIS and LIS, that is, IS have no obvious influence on those expected industry returns. Oil and gas companies are exposed to systematic asset price risks (Burbidge & Harrison, 1984; Gupta & Banerjee, 2019; Hamilton, 1983; Sadorsky, 1999) both oil prices and gasoline prices (Bacon, 1991; Deltas, 2008), and their investment behavior respond to the dynamic political and market environments (Mohn & Misund, 2011). Such results indicate that most global sentiments coefficients are positive, but *GIS* has an insignificant impact on industry returns. Notice that this is inconsistent with the findings of Chen et al. (2013). The effect of the IS is more significant on resource-based industries: basic materials (*MATS*) and industrials (*INDUS*) expected returns for those two industries' stocks could be significantly influenced by the fluctuations of global investors' emotion. One may conjecture that these stocks are more resistant because the investors may confirm their investment to avoid risks (Huang et al., 2014).

This finding implies that the world-wide optimism of stock investors could only slightly increase the expected industry returns, the negative impacts of GIS, that is, pessimism, is also limited.

5.2 | Local IS predict the SR but global sentiments do not

We find that local sentiments have heterogeneous effects on different industries' SR for emerging countries. The LIS significantly and positively affect the expected returns

TABLE 3 Correlation analysis between variables in this study

	MATS	GDS	SVS	FIN	HEA	INDUS	OIL	TECH	TELE	UTIL	GIS	LIS	BC
MATS	1												
GDS	0.7091	1											
SVS	0.7484	0.7477	1										
FIN	0.8208	0.6874	0.7140	1									
HEA	0.6120	0.6318	0.5907	0.5498	1								
INDUS	0.8686	0.7576	0.7552	0.8583	0.6161	1							
OIL	0.8157	0.6121	0.6647	0.7654	0.5647	0.8112	1						
TECH	0.5521	0.3644	0.3602	0.4731	0.3816	0.5363	0.4419	1					
TELE	0.7011	0.5459	0.6217	0.6890	0.4859	0.7150	0.7054	0.5158	1				
UTIL	0.7593	0.6043	0.6463	0.7710	0.5321	0.7772	0.7345	0.5288	0.6646	1			
GIS	0.1881	0.0640	0.1221	0.1170	0.0593	0.0975	0.0739	0.1219	0.1166	0.1431	1		
LIS	−0.0220	−0.0337	0.0619	0.0107	0.0121	0.0141	0.0082	−0.0536	−0.0729	−0.0451	0.0028	1	
BC	0.0679	−0.0320	0.0278	0.0038	0.0052	0.0076	0.1163	−0.0548	0.0362	0.0032	0.2420	0.1986	1

Note: The table shows the Pearson correlation coefficient value between 10 different industry sectors' stock returns, global investor sentiments, local investor sentiments and business cycle in the 2000 to 2014 period. All data involved are collected from DataStream.

TABLE 4 Regression coefficients results of fixed effect models

Variables	MATS	GDS	SVS	FIN	HEA
GIS	0.0579***	0.0125	0.0201	0.0244	−0.0089
	(0.015)	(0.015)	(0.022)	(0.013)	(0.011)
LIS	0.0564***	0.0350**	0.0334	0.0469***	0.0135
	(0.013)	(0.012)	(0.019)	(0.011)	(0.009)
BC	0.4043***	0.2016	0.2392	0.2364*	0.0774
	(0.112)	(0.104)	(0.162)	(0.097)	(0.085)
"constant"	−14.7369***	−6.0111**	−7.0518	−8.8560***	−0.7183
	(2.528)	(2.290)	(3.654)	(2.187)	(1.793)
R-square	0.041	0.013	0.005	0.022	0.004
VARIABLES	INDUS	OIL	TECH	TELE	UTIL
GIS	0.0264*	0.0026	0.0197	0.0207	0.0186
	(0.013)	(0.017)	(0.018)	(0.012)	(0.013)
LIS	0.0250*	0.0299	−0.0053	0.0188	0.0147
	(0.012)	(0.015)	(0.026)	(0.010)	(0.010)
BC	0.1997*	0.4211***	−0.1659	0.2009*	0.1550
	(0.099)	(0.122)	(0.140)	(0.088)	(0.091)
"constant"	−6.5200**	−6.1659*	−0.1938	−5.3780**	−4.3823*
	(2.224)	(2.831)	(3.258)	(2.002)	(2.047)
R-square	0.012	0.017	0.004	0.011	0.01

Note: This table reports the regression coefficient values of the fixed effect models for each explanatory variable (see text). The dependent variables are the expected returns for 10 different industry sectors stocks. Numbers in parentheses are the standard errors on the coefficients for the correspondent explanatory variables. The *, **, and *** indicate that the correspondent variable has passed the 95%, 99%, and 99.9% t-statistic test, respectively.

of only several specific industries. More precisely, for basic materials (*MATS*), consumer goods (*GDS*), financials (*FIN*) and industrials (*INDUS*), pessimistic LIS negatively influence the expected industry returns, while optimistic LIS produce positive impacts. Our research conclusions are consistent with Brown and Cliff (2005), Kumar and Lee (2006), Baker and Wurgler (2006, 2007), and Chen et al. (2013). In agreement with Baker and Wurgler (2006), in particular, optimistic and pessimistic LIS would lead to the overvaluation and undervaluation of stocks, respectively, and therefore increase and decrease the expected SR, respectively. Besides, we find that for many industries, such as health care (*HEA*), consumer services (*SVS*), oil and gas (*OIL*), technology (*TECH*), telecommunications (*TELE*), and utilities (*UTIL*), the expected industry returns have no significant correlation with LIS.

The corresponding interpretations vary with the mentioned industries. For health care (*HEA*) and telecommunications (*TELE*), two industries closely related to the interests of countries, many dramatic changes in those industries would bring a loss of national benefits. Hence, one can imagine that the regulators in these emerging economies try to keep those industries in control and prevent the industry returns from influences of IS. Of course, stocks of oil and gas (*OIL*) industry are regarded as traditional bond-like equities which have stable returns; thus, their expected returns are less influenced by IS.

One should stress that, for emerging countries, LIS are more influential than GIS. The interpretation goes as follows: because in the emerging stock markets, local investors could trade shares more freely than foreign investors, the expected industry returns could only demonstrate fluctuations induced by LIS. Interestingly, technology (*TECH*) is negatively (but not significantly) related to local sentiments; Baker and Wurgler (2006, 2007) provided some possible explanations which we consider to be realistic, for the phenomenon. They stated that those speculative stocks are difficult to price and thus hard to arbitrage; therefore, higher optimism periods of the stock market are expected to produce lower future returns of the stocks.

However, *GIS* cannot predict most of the industries' expected returns, while numerous existing studies, for example, Chen et al. (2013), suggest that global sentiments could significantly affect the expected return of several specific industries. This discrepancy between findings might be generated by the differences between research objects. In our research, we take emerging markets as the investigated object, rather than developed markets. Those emerging stock markets completely differ from the well-developed markets. On the one hand, the governments play significant roles in the stock markets of emerging countries; thus the stock markets' degrees of freedom are decreased (Claessens, Dasgupta, & Glen, 1995; Harvey, 1995). Take the Chinese stock market for example: the Chinese government has issued many regulations to limit capital investment from foreign country "for security reasons." Additionally, external investors are restricted to buy Chinese equities; therefore, changes of GIS could not bring tremendous impacts on emerging countries' expected SR. On the other hand, due to the limitations of investment knowledge, local stock traders might not respond notably or quickly to some foreign financial information; they are more sensitive to domestic news. Hence, the expected returns would not reflect the changes in the GIS. The foreign investor, however, may face formal or informal barriers when trading (Serra, 2000) because they might be more (or less) informative and efficient (Syamala & Wadhwa, 2019) but would also exposed to more risk of stock price crash (Vo, 2018; Zhou & Huang, 2019). The different types of investors, that is, individual and institutional investors may also exert different dynamic behavior due to the different momentums and horizons (Koesrindartoto, Aaron, Yusgiantoro, Dharma, & Arroisi, 2020). Due to such reasons, global sentiments could not be a significant explanatory variable for expected industry returns. This situation is thus inconsistent with the cases of developed stock markets, leading to the observed difference in behavior.

Notice that our empirical tests suggest that linear regression models of *LIS*, *GIS*, and *BC* factors might not significantly explain the fluctuations of expected industry SR. A possible explanation is that global sentiments are not a significant explanatory variable while local sentiments are not adequate to demonstrate all the changes of industry future returns. We suggest that a more complete anticipation of SR, for these markets, it seems to us, needs to bring other factors, such as the market status, company financial statements and risk-free rates, or even IPO news (Kim & Byun, 2010), into consideration. Moreover, other IS indexes might be invented (Bu & Pi, 2014) and thereafter tested. It would be also interesting to examine another type of causality, whether the stock market fluctuations, in such (emerging) countries and for this set of different industry sectors, influence the IS (Chen, Chiang, & So, 2003) rather than the contrary, as assumed here. Most likely the time correlations will have different spans; the investor, local or global, sentiment reaction might also have to be measured through other means, but the matter is another interesting open door.

6 | CONCLUSIONS

A survey of the literature implies that IS's influence on expected SR is an open field for research. Plenty of previous empirical analyses suggest that the fluctuations of IS can significantly sway future SR. However, we point out that only few studies investigate IS's heterogeneous effects on different industrial sectors' SR. Moreover, most of the existing research has focused on developed countries' stock markets as research object, leaving aside investigations of stock markets of emerging countries. Thus, in our study, we do analyze IS's (both local sentiments and global sentiments) heterogeneous effects on different industries' SR. To this aim, we follow Schmeling (2009) and Chen et al. (2013) research approaches and build a fixed effect linear regression model in order to study the correlations between IS and different industries' expected returns. Consequently, we selected six most significant emerging countries to represent the whole "developing world." Accordingly, the State Street Investor Confidence Index, each country (local) CCI, each country unemployment rate and the FTSE global price indexes are employed as the proxies for global sentiments, local sentiments, business cycle, and expected industry returns, respectively. After a group of statistical tests and necessary data adjustments, a panel data set has been constructed to perform an empirical analysis.

Sometimes confirming previously published conclusions about LIS's impacts, our empirical results also indicate some unique features of IS influencing mechanism for developing countries' stock markets. Unlike mature stock markets, the expected returns of emerging markets' stocks are insignificantly related to global sentiments; in other words, changes of global investors' emotions do not bring apparent changes of future SR. We conclude that likely due to government interventions and restrictions on foreign capital investment, emerging markets are insensitive to the impacts of global sentiment. For local sentiment-expected industry returns nexus, we find that local sentiments only positively and significantly affect the expected returns of basic materials (*MATS*), consumer goods (*GDS*), financials (*FIN*), and industrials (*INDUS*). Nevertheless, the other six industries' expected returns have no significant correlation with LIS. This finding, which is consistent with the general consensus in the former literature, confirms that IS have heterogeneous influences on different industries' SR. Moreover, the results show that it is inadequate to only use local sentiments, global sentiments and business cycle factor to anticipate future SR. We hold the opinion that other necessary factors, such as the market status, company financial statement, and risk-free rate, should be taken into account to conduct an effective anticipation of SR.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from Datastream. Restrictions apply to the availability of these data, which were used under license, at the University of Leicester Library, for this study. Data are available from the authors with the permission of University of Leicester and Datastream, as obtained from https://www2.le.ac.uk/library/find/databases/d/datastream.

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ENDNOTE

[1] An industry refers to a group of companies share the similar business activities, i.e., producing products, services or close substitutes. The classification schemes reflect different sources, revenues, and the stages of industry cycle, therefore respond differently to information.

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